

Empirical UX Evaluation: Preparation

Be prepared; that's the Boy Scouts' marching song... Don't be nervous, don't be flustered, don't be scared; be prepared!

— Tom Lehrer

Highlights

- Empirical UX evaluation plan
- Evaluation scope and rigor
- Goals for empirical UX evaluation
- Select team roles for empirical UX evaluation
- Prepare an effective range of user tasks
- Recruit participants
- Prepare for the session
- UX evaluation session work package
- Final pilot testing before evaluation

21.1 INTRODUCTION

21.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 21.1). This chapter is about how to prepare for empirical UX evaluation. Much of this will apply to planning for other kinds of evaluation, too.

21.1.2 Plan for the UX Evaluation Session

Empirical UX evaluation methods (see [Chapter 22](#)) entail data observed in performance by real user participants. The purpose of your plan for UX evaluation is to identify the most appropriate evaluation goals, methods, and activities for your project. If the plan will be read by people outside

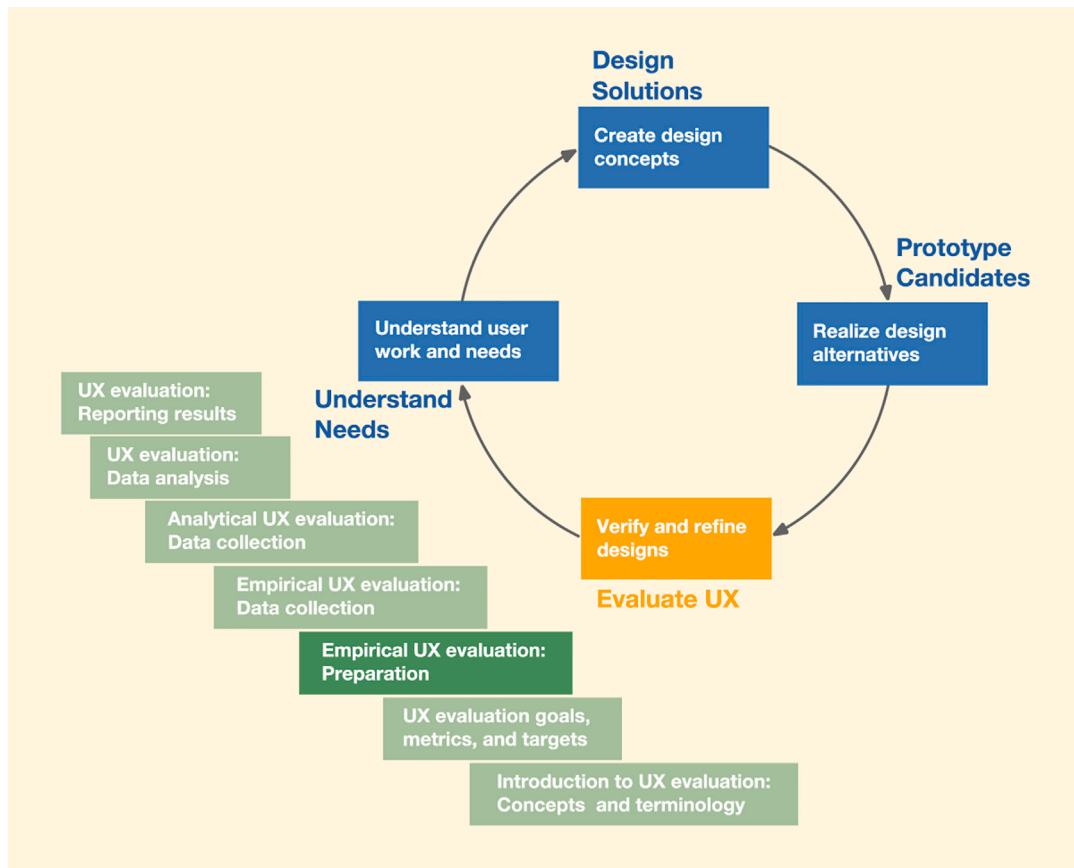


Fig. 21.1

You are here in the chapter on preparing for empirical evaluation within the evaluate UX activity in the context of the overall Wheel UX lifecycle.

your immediate project group, then you may want an upfront boilerplate introduction with some topics such as the following, described concisely:

- Overview of plan
- Overview of product or parts of the product being evaluated (for people outside the group)
- Goals for the user experience (what will make for a successful user experience)
- Description of the intended user population (user roles and targeted user classes)
- General evaluation methods and activities planned for this session
 - Description of your approach and techniques
- Estimated schedule
- Responsible personnel

21.2 EVALUATION SCOPE AND RIGOR

21.2.1 Evaluation Scope

You can perform UX evaluation at any scope. A large-scope approach is best for the early part of the agile UX funnel, where you would emphasize the ecological layer of the needs pyramid while evaluating the overall product or system architecture, user workflow, and conceptual design. A small-scope approach is best for frequent iterations within the sprints of the late agile UX funnel. Small-scope evaluation is used to address task-level design in the interaction level of the pyramid. As you may guess, you can address the emotional-needs layer of the pyramid at any scope.

21.2.2 Evaluation Rigor

You can also perform UX evaluation at any level of rigor. You can perform low-rigor evaluation in early design stages when the design is changing rapidly, and careful attention to preserving detail in the data would be wasted. You may also have to perform your UX evaluation at a relatively low level of rigor in the late funnel simply because of the pressure to keep up with the agile sprints.

Larger organizations once devoted huge resources to do rigorous UX evaluation with careful attention to detail and full preservation of the data—both data purity and data completeness. Rigorous methods still have a place in projects that demand it; however, with the shift of attention toward agile methods, it is more difficult to justify the time and cost of high rigor in most projects.

21.3 GOALS FOR AN EMPIRICAL UX EVALUATION SESSION

One of the first things to do in an evaluation plan is to set and prioritize your evaluation goals specific to this project. Identify the most important design issues and user tasks to investigate. Decide which parts of the system or functionality you simply will not have time to view.

21.4 SELECT TEAM ROLES

21.4.1 Participation and Buy-in

Encourage your whole project team to participate in at least some evaluation. Broad participation begets buy-in and ownership, necessary for your results to be taken as a serious mandate to fix problems. Some specific roles are described here. Anyone on the team can learn as an observer.

21.4.2 Facilitator

Your facilitator is the leader of the evaluation team, the one who makes sure it all works successfully. The facilitator has the primary responsibility for planning and executing the evaluation sessions.

21.4.3 Prototype Executor

If you are using a low-fidelity click-through wireframe prototype, you need to select a prototype executor, a person to execute the prototype and move it through its paces as users interact.

21.4.4 Quantitative Data Collectors

If you plan to include quantitative data collection, you will need people to do it. Depending on your UX metrics and quantitative data collection instruments, people designated to collect quantitative data may be there with stopwatches and counters.

21.4.5 Qualitative Data Collectors

Select as many team members as possible to serve as qualitative data collectors and recorders. Everyone should be ready to collect qualitative data, especially critical incident data (see [Section 22.3.1](#)).

21.5 PREPARE AN EFFECTIVE RANGE OF USER TASKS AND APPROPRIATE SURVEY INSTRUMENTS

21.5.1 Benchmark Task Descriptions

Benchmark tasks (see [Section 20.5.1](#)) portray representative, frequent, and critical tasks that apply to the key work role and user class represented by each participant. If you have specified one or more benchmark tasks to generate quantitative measures, you should now have the corresponding task descriptions and UX target metrics (see [Sections 20.5.1.1 and 20.8](#)) ready to guide your data collection and to compare with observed results.

Each benchmark task description should be printed and ready to use by participants to generate data to be measured. Make sure each task description says only what to do, with no hints about how to do it. Avoid language that telegraphs any part of the design (e.g., names of user interface objects or user actions, or words from labels or menus). Print each task on a separate sheet of paper so users can focus on just one task at a time. For more on survey instruments to generate quantitative measures, see [Section 22.2.2](#).

21.5.2 Other Kinds of Task Descriptions

In addition to benchmark tasks, consider some of these other kinds of task descriptions.

Unmeasured tasks. Unmeasured tasks are user tasks for which participant performance will not be measured quantitatively. Evaluators can use these representative tasks to add breadth to qualitative evaluation by addressing aspects of the design not covered in some way by the benchmark tasks.

Exploratory free use tasks. Sometimes it is useful to observe the participant in informal interaction with the design, a free-play period without the constraints of predefined tasks. This does not necessarily mean that they are doing tasks, maybe just exploring.

User-defined tasks. Sometimes tasks that users come up with will address unexpected aspects of your design (Cordes, 2001). As part of an evaluation session, ask each participant extemporaneously to come up with tasks to try.

21.6 RECRUIT PARTICIPANTS

The next step in preparing for empirical UX evaluation is selection and recruitment of participants—finding representative users, usually outside your team and often outside your project organization, to help with evaluation.

21.6.1 Establish Budget and Schedule for Recruiting User Participants Upfront

Finding and recruiting evaluation participants may be part of the process where you are tempted to cut corners and save a little on the budget or may be something you think to do at the last minute. To protect the larger investment already made in the UX lifecycle process and in setting up formative evaluation so far, you need to secure a reasonable amount of resources (e.g., money in the budget to pay the participants and time in the schedule to recruit the full range and number of evaluation participants you will need). If you do this kind of evaluation infrequently, you can engage the services of a professional recruiter to do your participant recruiting or even a UX evaluation consulting group to do the whole evaluation.

21.6.2 Identify the Right Kinds of Participants

Look for participants who are representative users (i.e., participants who match your target work role's user class descriptions). If you have multiple work roles and user classes, you should try to recruit participants representing each category.

21.6.3 Determine the Right Number of Participants

The question of how many participants you need is dependent on the kind of evaluation you are doing and the conditions under which you are doing it. There are some rules of thumb, such as the famous “three to five participants are enough” maxim, which is quoted so often out of context as to be almost meaningless.

The good news is that your experience and intuition will be effective touchstones for knowing when you have gotten the most out of an iteration of UX evaluation and when to move on. More often it is about getting in, getting some insight, and getting out.

21.6.4 Consider Recruiting Methods and Screening

Now the question arises as to where to find participants. If you are working with a customer organization, inform your customer early on about how your evaluation process will proceed so you will have the best chance of getting representative users from the customer organization at appropriate times. Only as a last resort, look at coworkers and colleagues.

Following are hints for successful participant recruiting:

- Newspaper ads and emails can work to recruit participants, but these methods are usually inefficient.
- If the average person off the street fits your participant profile (e.g., for a consumer software application), hand out leaflets in shopping malls and parking lots or post notices in grocery stores or in other public places (e.g., libraries).
- For domain-complex systems, typically there exist specialized sales roles who, as a last resort, can act as user proxies. They are most familiar with market and customer needs because of the high-touch interactions they have with customers in these domains.

21.6.5 Decide on Incentives and Remuneration

Generally, you should offer your participants some kind of remuneration. You will usually pay a modest hourly fee or you can offer various kinds of premium gifts, such as coffee mugs, gift certificates, T-shirts, or free pizza.

21.6.6 Recruit for Codiscovery Evaluation

Consider recruiting pairs of participants specifically for codiscovery evaluation, a qualitative data collection technique employing two or more participants interacting in a team approach to evaluation, usually with a think-aloud

data collection technique (see [Section 22.3.2](#)). Two people can verbalize more naturally, yielding multiple viewpoints expressed within conversational interplay. Your goal is to find people who will work well together during evaluation and, as a practical matter, who are available at the same time.

21.6.7 Select Participants for Subsequent Iterations

Sometimes reusing participants can make sense. This way, you can get a reaction to design changes from the previous cycle in addition to a new set of data on the modified design from the new participants. Calling on a previously used participant tells them you value their help and gives them a kind of empowerment, a feeling that they are helping to make a difference in your design.

21.7 PREPARE FOR THE SESSION

21.7.1 Lab and Equipment

If you are planning lab-based evaluation, the most obvious aspect of preparation is to have the lab available and configured for your needs. If you plan to collect quantitative UX data, prepare by having on hand the right kind of timers for tasks and counters for errors, from simple stopwatches to instrumented software for automatically extracting timing data.

Example: A Modern UX Lab at Bloomberg LP

Bloomberg LP, a leader in financial informatics, employs a modern UX evaluation lab with two areas—a participant room and an observation room—each with an independent entrance and separated by a one-way mirror. The participant room has a multi-monitor workstation ([Fig. 21.2](#)) on which Bloomberg’s desktop applications are evaluated.

In the participant room, there is also an area designated for evaluations with low-fidelity paper prototypes ([Fig. 21.3](#)) or mobile devices ([Fig. 21.4](#)).

[Fig. 21.5](#) provide a view of the observation room, which is kept dark to prevent people in the participant room from seeing through. In this image, you can see the participant room showing through the one-way mirror. The lab is set up to stream up to five selections of the seven video sources and four screen capture sources from the participant room to the large screens seen at the top in the observation room. This lab has been instrumental in defining the UX designs of Bloomberg’s flagship desktop and mobile applications.

Fig. 21.2

Desktop evaluation in the Bloomberg UX evaluation lab (courtesy of Shawn Edwards, Fahd Arshad, Pam Snook, and Vera Newhouse, Bloomberg LP.).



Fig. 21.3

Paper prototype evaluation in the Bloomberg UX evaluation lab (courtesy of Shawn Edwards, Fahd Arshad, Pam Snook, and Vera Newhouse, Bloomberg LP.).



21.7.2 Session Parameters

Evaluators must determine protocol and procedures for conducting the evaluation—exactly what will happen and for how long during an evaluation session with a participant.

Task and session lengths. The typical length of time of an evaluation session for one participant is roughly 30 to 120 minutes; however, it is possible that a real-world UX evaluation session can become a day-long experience for a participant. The idea is to get as much as possible from each user without causing burnout.



Fig. 21.4

Mobile device evaluation in the Bloomberg UX evaluation lab (courtesy of Shawn Edwards, Fahd Arshad, Pam Snook, and Vera Newhouse, Bloomberg LP.).



Fig. 21.5

The observation room in the Bloomberg UX evaluation lab (courtesy of Shawn Edwards, Fahd Arshad, Pam Snook, and Vera Newhouse, Bloomberg LP.).

Number of full lifecycle iterations. As a loose rule of thumb from our experience, an ideal number of full UX cycle iterations per version or release is about three, but resource constraints often limit it to fewer. In many

projects, you can expect only one iteration. Of course, any iterations are better than none.

21.7.3 Informed Consent

When we collect empirical data involving human subjects, we have certain legal and ethical responsibilities, even though there is very little risk of a participant being harmed in UX evaluation.

We still have professional obligations, which center on the informed consent form, a document that, when signed, gives you permission to use the data gathered. This document also establishes the rights of your participants and serves as legal protection for you and your organization.

Example: Simple Informed Consent Form

Informed Consent for Participant of Development Project

<Name of your development organization> <Date or version number of form>

Title of Project: <Project title>

Project team member(s) directly involved: <Team member names>

Project manager: <Project manager name>

I. THE PURPOSE OF YOUR PARTICIPATION IN THIS PROJECT

As part of the <project title> project, you are invited to participate in evaluating and improving various designs of <name of system or product>, <description of system or product>.

II. PROCEDURES

You will be asked to perform a set of tasks using the <name of system or product>. These tasks consist of <description of range of tasks>. Your role in these tests is to help us evaluate the designs. We are not evaluating you or your performance in any way. As you perform various tasks with the system, your actions and comments will be noted, and you will be asked to describe verbally your learning process. You may be asked questions during and after the evaluation to clarify our understanding of your evaluation. You may also be asked to fill out a questionnaire relating to your usage of the system. The evaluation session will last no more than 4 hours, with the typical session being about 2 hours. The tasks are not very tiring, but you are welcome to take rest breaks as needed. If you prefer, the session may be divided into two shorter sessions.

III. RISKS

There are no known risks to the participants of this study.

IV. BENEFITS OF THIS PROJECT

Your participation in this project will provide information that may be used to improve our designs for <name of system or product>.

No guarantee of further benefits has been made to encourage you to participate. (Change this if a benefit such as a payment or a gift is offered.) You are requested to refrain from discussing the evaluation with other people who may be in the candidate pool from which other participants are drawn.

V. EXTENT OF ANONYMITY AND CONFIDENTIALITY

The results of this study will be kept strictly confidential. Your written consent is required for the researchers to release any data identified with you as an individual. The information you provide will have your name removed, and only a subject number will identify you during analyses and any written reports of the research.

The session may be recorded. If it is recorded, the recordings will be stored securely, viewed only by the project team members, and erased after 3 months. If team members wish to use a portion of your recording for any other purpose, they will get your written permission before using it. Your signature on this form does not give them permission to show your recording to anyone else.

VI. COMPENSATION

Your participation is voluntary and unpaid. (Change this if a benefit, such as a payment or a gift, is offered.)

VII. FREEDOM TO WITHDRAW

You are free to withdraw from this study at any time for any reason.

VIII. APPROVAL OF RESEARCH

This research has been approved, as required, by the Institutional Review Board <or the name of your review committee> for projects involving human subjects at <your organization>.

IX. PARTICIPANT RESPONSIBILITIES AND PERMISSION

I voluntarily agree to participate in this study, and I know of no reason I cannot participate. I have read and understand the informed consent and conditions of this project. I have had all my questions answered. I hereby acknowledge the above and give my voluntary consent for participation in this project. If I participate, I may withdraw at any time without penalty. I agree to abide by the rules of this project.

Signature:

Date:

Name (please print):

Contact: phone or email:

21.7.4 Other Paperwork

User instructions include the following:

- Explain briefly the purpose of the evaluation.
- Tell a little bit about the system the participant will be using.
- Describe what the participant will be expected to do and the procedure to be followed by the participant.
- Make it clear that you are not evaluating the participant: “You are helping us evaluate the system—we are not evaluating you!”
- Guarantee confidentiality with respect to individual information and anonymity of data.

Print out and copy the general instructions so that you can give one to each participant.

21.7.4.1 *Non-disclosure agreement*

Sometimes a non-disclosure agreement (NDA) is required by you or your customer organization to protect the intellectual property contained in the design. If you have an NDA, print out copies for reading, signing, and sharing with participants.

21.7.4.2 *Questionnaires and surveys*

If your evaluation plan includes administration of one or more participant questionnaires, then be sure you have a good supply available. It is best to keep blank questionnaires in the control room or away from where a newly arriving participant could read them in advance.

Exercise 21.1 Empirical UX Evaluation Preparation for Your Product or System

Goal: To practice preparing for a simple empirical evaluation (see [Section 5.6.1.2](#)).

Activities: If you are working with a team, get together for this activity.

- Decide roles for team members. Include at least a facilitator and a prototype executor, plus a quantitative data recorder and one or more critical incident recorders.
- In addition, if you are doing this exercise in a classroom with other teams, assign two team members as participants to trade to another team when you start data collection in the next exercise.

- The prototype executor should get out the wireframe prototype deck (or other prototype) you made in a previous exercise and become familiar with the navigation.
- This activity works well for a team of about four. If you have more or fewer members in your team, it is easy to make adjustments. If there are only two of you, for example, one person can be the executor and the other person can record critical incidents and time the benchmark tasks. If there are four or five of you, the extra people will be valuable in helping record critical incidents. If you have been working alone on all the previous exercises, you may want to find a couple of other people to help you run the evaluation. In addition, and in any case, you need to recruit two people to serve as participants to evaluate your prototype.
- Get out the UX target table you made in a previous exercise.
- Have the benchmark tasks that you created in a previous exercise, each printed on a separate piece of paper.
- Assuming you used a questionnaire for subjective data in your evaluation session, get out copies of the questionnaire, one for each participant you will be using, and circle the questions you want participants to answer.
- Review your evaluation protocols.

Deliverables: Have everything ready for the next exercise, data collection.

Schedule: It should not take too long to get ready for evaluation.

21.7.5 Do Final Pilot Testing: Fix Your Wobbly Wheels

If your UX evaluation plan involves using a prototype, give it a final shakedown to be sure it is robust enough to support evaluation without breaking. This step applies to any level of fidelity if your prototype will be seen by people beyond your UX team.

21.8 SEGUE

In this chapter, we discussed preparation for empirical evaluation. In the next chapter, we get into conducting the evaluation.